20261110 Quiz

for (MATH230) Probability & Statistics

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Exercise 7.16. Let X have a gamma distribution with shape $\alpha > 0$ and scale $\theta > 0$, with density

$$f_X(x) = \frac{x^{\alpha-1}e^{-x/\theta}}{\Gamma(\alpha)\theta^\alpha}, \qquad x > 0.$$

Show that the rth moment about the origin is $\mu'_r = \theta^r\Gamma(\alpha+r)/\Gamma(\alpha)$ for any nonnegative integer r. *Hint.* Substitute $u = x/\theta$.

Solution. From the definition of a moment and the given density,

$$\begin{aligned}\mu'_r = \mathbb{E}[X^r] &= \frac{1}{\Gamma(\alpha)\theta^\alpha}\int_0^\infty x^{\alpha+r-1}e^{-x/\theta}\,\mathrm{d}x \\ &= \frac{\theta^r}{\Gamma(\alpha)}\int_0^\infty u^{\alpha+r-1}e^{-u}\,\mathrm{d}u = \boxed{\frac{\theta^r\Gamma(\alpha+r)}{\Gamma(\alpha)}}.\end{aligned}$$

Here $x = \theta u$ and $\mathrm{d}x = \theta\,\mathrm{d}u$; the last integral is the gamma function. □

Exercise 7.17. The random variable X has the discrete uniform distribution

$$\mathbb{P}(X = x) = \frac{1}{k}, \qquad x = 1, 2, \ldots, k,$$

where $k \in \mathbb{N}$. Show that its moment-generating function is

$$M_X(t) = \frac{e^t(1-e^{kt})}{k(1-e^t)} \quad (t \neq 0).$$

Solution. The finite geometric-series formula gives, for $t \neq 0$,

$$M_X(t) = \mathbb{E}[e^{tX}] = \frac{1}{k}\sum_{x=1}^{k} e^{tx} = \frac{e^t}{k}\sum_{j=0}^{k-1}(e^t)^j = \boxed{\frac{e^t(1-e^{kt})}{k(1-e^t)}}.$$

At $t = 0$ the displayed quotient has the form $0/0$, but the defining sum gives $\boxed{M_X(0) = 1}$; this also equals the limit of the quotient as $t \to 0$. □

Exercise 7.18. Let $X \sim \mathrm{Geom}(p)$, with $0 < p < 1$ and $q = 1 - p$, so that $\mathbb{P}(X = x) = pq^{x-1}$ for $x = 1, 2, \ldots$. Show that

$$M_X(t) = \frac{pe^t}{1-qe^t}, \qquad t < -\ln q,$$

and use the MGF to find the mean and variance of X.

Solution. For a convergent geometric series we require $qe^t < 1$, equivalently $t < -\ln q$. Hence

$$M_X(t) = \sum_{x=1}^{\infty} e^{tx} pq^{x-1} = pe^t \sum_{j=0}^{\infty} (qe^t)^j = \boxed{\frac{pe^t}{1-qe^t}}.$$

Differentiating in the interval of convergence gives

$$M_X'(t) = \frac{pe^t}{(1-qe^t)^2}, \qquad M_X''(t) = \frac{pe^t(1+qe^t)}{(1-qe^t)^3}.$$

Since $1 - q = p$,

$$\mathbb{E}[X] = M_X'(0) = \boxed{\frac{1}{p}}, \qquad \mathrm{Var}(X) = M_X''(0) - M_X'(0)^2 = \frac{1+q}{p^2} - \frac{1}{p^2} = \boxed{\frac{q}{p^2}}.$$

□

Exercise 7.19. Let $X \sim \mathrm{Pois}(\mu)$, with $\mu > 0$, so that $\mathbb{P}(X = x) = e^{-\mu}\mu^x/x!$ for $x = 0, 1, 2, \ldots$. Show that $M_X(t) = e^{\mu(e^t-1)}$, and use the MGF to find the mean and variance of X.

Solution. Using the exponential series,

$$M_X(t) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\mu}\mu^x}{x!} = e^{-\mu} \sum_{x=0}^{\infty} \frac{(\mu e^t)^x}{x!} = \boxed{e^{\mu(e^t-1)}}.$$

The MGF is finite for every $t \in \mathbb{R}$, and

$$M_X'(t) = \mu e^t M_X(t), \qquad M_X''(t) = (\mu e^t + \mu^2 e^{2t}) M_X(t).$$

As $M_X(0) = 1$,

$$\mathbb{E}[X] = M_X'(0) = \boxed{\mu}, \qquad \mathrm{Var}(X) = M_X''(0) - M_X'(0)^2 = \mu + \mu^2 - \mu^2 = \boxed{\mu}.$$

□

Exercise 7.20. The MGF of a certain Poisson random variable X is $M_X(t) = e^{9(e^t-1)}$. Find $\mathbb{P}(\mu - \sigma \le X \le \mu + \sigma)$, where $\mu = \mathbb{E}[X]$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.

Solution. Comparing with the Poisson MGF identifies $X \sim \mathrm{Pois}(9)$, hence $\mu = 9$ and $\sigma = 3$. Since X is integer-valued,

$$\begin{aligned}\mathbb{P}(\mu - \sigma \le X \le \mu + \sigma) &= \mathbb{P}(6 \le X \le 12) \\ &= e^{-9} \sum_{x=6}^{12} \frac{9^x}{x!} \approx \boxed{0.7601}.\end{aligned}$$

□

Exercise 7.21. Show that the MGF of a chi-squared random variable $X \sim \chi^2_\nu$ with ν degrees of freedom is

$$M_X(t) = (1-2t)^{-\nu/2}.$$

Solution. Its density is $f_X(x) = x^{\nu/2-1}e^{-x/2}/(2^{\nu/2}\Gamma(\nu/2))$ for $x > 0$. Thus, for $t < 1/2$,

$$\begin{aligned} M_X(t) &= \frac{1}{2^{\nu/2}\Gamma(\nu/2)} \int_0^\infty x^{\nu/2-1} e^{-(1/2-t)x} \, \mathrm{d}x \\ &= \frac{\Gamma(\nu/2)}{2^{\nu/2}\Gamma(\nu/2)(1/2-t)^{\nu/2}} = \boxed{(1-2t)^{-\nu/2}}. \end{aligned}$$

The gamma integral is finite precisely when $1/2 - t > 0$; this is the domain of the MGF. □

Exercise 7.22. The MGF of a chi-squared random variable X with ν degrees of freedom is $M_X(t) = (1-2t)^{-\nu/2}$ for $t < 1/2$. Use it to show that $\mathbb{E}[X] = \nu$ and $\mathrm{Var}(X) = 2\nu$.

Solution. Differentiation gives

$$M_X'(t) = \nu(1-2t)^{-\nu/2-1}, \qquad M_X''(t) = \nu(\nu+2)(1-2t)^{-\nu/2-2}.$$

Therefore

$$\mathbb{E}[X] = M_X'(0) = \boxed{\nu}, \qquad \mathrm{Var}(X) = M_X''(0) - M_X'(0)^2 = \nu(\nu+2) - \nu^2 = \boxed{2\nu}.$$

□

Exercise 7.24. Let X be a continuous random variable with density f_X. Suppose its MGF is finite on an open interval containing 0. Recall that

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) \, \mathrm{d}x.$$

Expand e^{tx} in a Maclaurin series and integrate term by term to show that

$$M_X(t) = 1 + \mu t + \mu_2' \frac{t^2}{2!} + \cdots + \mu_r' \frac{t^r}{r!} + \cdots,$$

where $\mu_r' = \mathbb{E}[X^r]$ and $\mu = \mathbb{E}[X]$.

Solution. For t sufficiently close to 0, the MGF hypothesis permits term-by-term integration of the exponential series:

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} \left(\sum_{r=0}^{\infty} \frac{(tx)^r}{r!} \right) f_X(x) \, \mathrm{d}x \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!} \int_{-\infty}^{\infty} x^r f_X(x) \, \mathrm{d}x = \sum_{r=0}^{\infty} \frac{\mu_r'}{r!} t^r. \end{aligned}$$

Since

$$\mu_0' = \int_{-\infty}^{\infty} f_X(x) \, \mathrm{d}x = 1, \qquad \mu_1' = \mu,$$

this is the required expansion. Equivalently, $M_X^{(r)}(0) = \mu_r'$. □

Exercise 8.1. Define a suitable population for each of the following samples.

(a) Customers contacted by a car manufacturer for feedback after their vehicles were serviced at its authorized service center.

(b) Ten randomly selected college students, of whom four are girls.

(c) Total marks collected from 100 graduate students at a university.

(d) Fifty observations made to determine the lifetime of a particular brand of truck tire.

Solution.

(a) The population is all customers whose vehicles are serviced at the manufacturer's authorized service center (in the period of interest).

(b) The population is all students at the college from which the ten were randomly selected, classified by gender.

(c) The population is the set of graduate students at that university, with each student's total marks as the measured characteristic.

(d) The population is the lifetimes of all tires of that brand under the specified usage conditions.

□

Exercise 8.2. The waiting times, in minutes, of ten patients at a doctor's office are

$$5,\ 11,\ 9,\ 5,\ 10,\ 15,\ 6,\ 10,\ 5,\ 10.$$

Treat these as a random sample and find

(a) the mean.

(b) the median.

(c) the mode.

Solution. The ordered data are $5, 5, 5, 6, 9, 10, 10, 10, 11, 15$, with $n = 10$ and total 86.

(a) $\bar{x} = \dfrac{86}{10} = \boxed{8.6 \text{ min}}$.

(b) The middle values are 9 and 10, so $\tilde{x} = \dfrac{9 + 10}{2} = \boxed{9.5 \text{ min}}$.

(c) Both 5 and 10 occur three times, the greatest frequency. The modes are $\boxed{5 \text{ and } 10 \text{ min}}$.

□

Exercise 8.3. Reaction times, in seconds, for nine subjects given a stimulant were recorded as

$$2.5,\ 3.6,\ 3.1,\ 4.3,\ 2.9,\ 2.3,\ 2.6,\ 4.1,\ 3.4.$$

Calculate

(a) the mean.

(b) the median.

Solution.

(a) Since $\sum x_i = 28.8$, $\bar{x} = \dfrac{28.8}{9} = \boxed{3.2 \text{ s}}$.

(b) The ordered values are $2.3, 2.5, 2.6, 2.9, 3.1, 3.4, 3.6, 4.1, 4.3$. The fifth is the middle observation, so $\boxed{\tilde{x} = 3.1 \text{ s}}$.

□

Exercise 8.5. The numbers of incorrect answers on a true-false competency test for a random sample of 15 students are

$$2,\ 1,\ 3,\ 0,\ 1,\ 3,\ 6,\ 0,\ 3,\ 3,\ 5,\ 2,\ 1,\ 4,\ 2.$$

Find

(a) the mean.
(b) the median.
(c) the mode.

Solution. The sum is 36. In ascending order the data are $0, 0, 1, 1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 5, 6$.

(a) $\bar{x} = \dfrac{36}{15} = \boxed{2.4}$.
(b) The eighth of 15 ordered values is 2, so $\boxed{\tilde{x} = 2}$.
(c) The value 3 occurs four times, more than any other value. Thus the mode is $\boxed{3}$.

□

Exercise 8.6. The numbers of medical leaves claimed in a year by ten employees are

$$5,\ 12,\ 3,\ 7,\ 11,\ 1,\ 85,\ 5,\ 9,\ 5.$$

Find the mean, median, and mode. Which gives the best measure of the center? Explain.

Solution. The observations sum to 143, and their ordered list is $1, 3, 5, 5, 5, 7, 9, 11, 12, 85$. Therefore

$$\bar{x} = \boxed{14.3}, \qquad \tilde{x} = \frac{5+7}{2} = \boxed{6}, \qquad \text{mode} = \boxed{5}.$$

The unusually large observation 85 raises the mean far above the bulk of the data. The *median* 6 is the preferable measure of center here because it is resistant to this extreme value. □

Exercise 8.7. The examination scores of twenty students selected from a city school are

$$45,\ 57,\ 68,\ 34,\ 50,\ 32,\ 89,\ 47,\ 97,\ 67,$$
$$79,\ 84,\ 43,\ 35,\ 68,\ 55,\ 72,\ 63,\ 68,\ 49.$$

Calculate

(a) the mean.
(b) the mode.

Solution.

(a) The score total is 1202, so $\bar{x} = \dfrac{1202}{20} = \boxed{60.1}$.
(b) The score 68 appears three times; all other scores appear once. The mode is $\boxed{68}$.

□

Exercise 8.8. Phosphates in household detergents can contribute to water pollution. The following are phosphate amounts (grams per prescribed load) for twelve sampled detergents:

Detergent	Grams	Detergent	Grams
A & P Blue Sail	48	Ajax	31
Dash	47	Sears	30
Concentrated All	42	Fab	29
Cold Water All	42	Cold Power	29
Breeze	41	Bold	29
Oxydol	34	Rinso	26

For the phosphate data, find

(a) the mean.
(b) the median.
(c) the mode.

Solution. There are 12 observations, with total 428. In ascending order the values are

$$26,\ 29,\ 29,\ 29,\ 30,\ 31,\ 34,\ 41,\ 42,\ 42,\ 47,\ 48.$$

(a) $\bar{x} = \dfrac{428}{12} = \boxed{\dfrac{107}{3} \approx 35.67 \text{ g}}$.

(b) The middle values are 31 and 34, so $\tilde{x} = \dfrac{31+34}{2} = \boxed{32.5 \text{ g}}$.

(c) The most frequent amount is 29 (three occurrences). The mode is $\boxed{29 \text{ g}}$.

□

Exercise 8.9. The waiting times, in minutes, of ten patients are $5, 11, 9, 5, 10, 15, 6, 10, 5, 10$. Treat them as a random sample and find

(a) the range.
(b) the sample standard deviation.

Solution. (a) The smallest and largest times are 5 and 15, so $\boxed{r = 15 - 5 = 10 \text{ min}}$.
(b) For $n = 10$, the sum of observations is 86 and the sum of squares is 838. The computational formula gives

$$s^2 = \frac{838 - 86^2/10}{10 - 1} = \frac{164}{15} \approx 10.9333 \text{ min}^2, \qquad \boxed{s = \sqrt{\frac{164}{15}} \approx 3.307 \text{ min}}.$$

□

Exercise 8.10. For the reaction times (seconds)

$$2.5,\ 3.6,\ 3.1,\ 4.3,\ 2.9,\ 2.3,\ 2.6,\ 4.1,\ 3.4,$$

calculate

(a) the range.

(b) the sample variance using the definition $s^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$.

Solution. (a) $\boxed{r = 4.3 - 2.3 = 2.0 \text{ s}}$.
(b) The sample mean is $\bar{x} = 3.2$. By the definition,

$$s^2 = \frac{1}{8}\sum_{i=1}^{9}(x_i - \bar{x})^2 = \frac{1}{8}\sum_{i=1}^{9}(x_i - 3.2)^2 = \frac{3.98}{8} = \boxed{0.4975 \text{ s}^2}.$$

□

Exercise 8.11. For the sample of numbers of incorrect test answers

$$2,\ 1,\ 3,\ 0,\ 1,\ 3,\ 6,\ 0,\ 3,\ 3,\ 5,\ 2,\ 1,\ 4,\ 2,$$

calculate the sample variance using

(a) the definition $s^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$.

(b) the computational formula of Theorem 8.1, $s^2 = \dfrac{n\sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}{n(n-1)}$.

Solution. Here $n = 15$, $\bar{x} = 2.4$, and the data give

$$\sum_{i=1}^{15} x_i = 36, \qquad \sum_{i=1}^{15} x_i^2 = 128.$$

(a) By the definition, group the equal observations:

$$\begin{aligned}\sum_{i=1}^{15}(x_i - \bar{x})^2 &= 2(0-2.4)^2 + 3(1-2.4)^2 + 3(2-2.4)^2 \\ &\quad + 4(3-2.4)^2 + (4-2.4)^2 + (5-2.4)^2 + (6-2.4)^2 \\ &= 41.6.\end{aligned}$$

Therefore

$$s^2 = \frac{41.6}{14} = \boxed{\frac{104}{35} \approx 2.9714}.$$

(b) By Theorem 8.1,

$$\begin{aligned}s^2 &= \frac{n\sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}{n(n-1)} \\ &= \frac{15(128) - 36^2}{15(14)} = \boxed{\frac{104}{35} \approx 2.9714}.\end{aligned}$$

□

Exercise 8.12. The saturated-fat contents, in grams, of eight randomly selected cereal bars are

$$0.65,\ 0.72,\ 0.45,\ 0.55,\ 0.58,\ 0.39,\ 0.68,\ 0.52.$$

Calculate

(a) the sample mean.

(b) the sample variance.

Solution. For $n = 8$, the sum of observations is 4.54 and the sum of squares is 2.6672.

(a) $\bar{x} = \dfrac{4.54}{8} = \boxed{0.5675 \text{ g}}$.

(b) The computational formula yields

$$s^2 = \frac{2.6672 - (4.54)^2/8}{7} = \boxed{0.0129643 \text{ g}^2}.$$

□

Exercise 8.13. The heights, in meters, of twenty randomly selected college seniors are

1.53	1.48	1.64	1.70	1.56
1.79	1.69	1.74	1.79	1.73
1.43	1.68	1.64	1.75	1.76
1.59	1.68	1.74	1.73	1.71

Calculate the sample standard deviation.

Solution. For $n = 20$, the data give $\sum x_i = 33.36$ and $\sum x_i^2 = 55.8414$, so $\bar{x} = 1.668$ m. The computational formula gives

$$s^2 = \frac{55.8414 - (33.36)^2/20}{19} \approx 0.01036421 \text{ m}^2,$$
$$s = \boxed{\sqrt{s^2} \approx 0.1018 \text{ m}}.$$

□

Exercise 8.14. For a sample $x_1, \ldots, x_n$ with $n \geq 2$, prove that

(a) adding or subtracting a constant c from every observation leaves the sample variance unchanged.

(b) multiplying every observation by a constant c multiplies the sample variance by c^2.

Solution. Write

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

(a) If $y_i = x_i + c$, then $\bar{y} = \bar{x} + c$, so $y_i - \bar{y} = x_i - \bar{x}$. Thus $\boxed{s_y^2 = s_x^2}$. Replacing c by $-c$ proves the subtraction case.

(b) If $y_i = cx_i$, then $\bar{y} = c\bar{x}$ and $y_i - \bar{y} = c(x_i - \bar{x})$. Consequently,

$$s_y^2 = \frac{1}{n-1} \sum_{i=1}^{n} c^2 (x_i - \bar{x})^2 = \boxed{c^2 s_x^2}.$$

□

Exercise 8.15. First calculate the sample variance of $8, 10, 16, 18, 24, 25$. Then use the variance rules for translation and scaling to find the variances of

(a) $16, 20, 32, 36, 48, 50$.

(b) $12, 14, 20, 22, 28, 29$.

Solution. For the original sample, $n = 6$, $\sum x_i = 101$, and $\sum x_i^2 = 1945$. Hence

$$s_x^2 = \frac{1945 - 101^2/6}{5} = \boxed{\frac{1469}{30} \approx 48.9667}.$$

(a) Each value is doubled, so $\boxed{s_{2x}^2 = 4s_x^2 = \frac{2938}{15} \approx 195.8667}$.

(b) Each value is increased by 4, so $\boxed{s_{x+4}^2 = s_x^2 = \frac{1469}{30} \approx 48.9667}$.

□

Exercise 8.16. In twelve one-day cricket matches during the 2014–15 season, a university team captain scored

$$84,\ 25,\ 74,\ 53,\ 40,\ 31,\ 64,\ 71,\ 18,\ 63,\ 88,\ 49.$$

Find

(a) the mean score.

(b) the median score.

Solution.

(a) The total is 660, so $\bar{x} = \frac{660}{12} = \boxed{55}$ runs.

(b) The ordered scores are $18, 25, 31, 40, 49, 53, 63, 64, 71, 74, 84, 88$. The sixth and seventh are 53 and 63, so $\tilde{x} = \frac{53 + 63}{2} = \boxed{58}$ runs.

□

Exercise 8.17. All possible samples of size 25 are drawn from a normal population with mean 60 and standard deviation 9. Find the probability that the sample mean $\bar{X}$ lies between $\mu_{\bar{X}} - 0.96\sigma_{\bar{X}}$ and $\mu_{\bar{X}} + 0.25\sigma_{\bar{X}}$. Assume the sample means can be measured to any degree of accuracy.

Solution. Since the population is normal, $\bar{X} \sim \mathcal{N}(60, 9^2/25)$ exactly; thus $\sigma_{\bar{X}} = 9/5$. Standardization gives

$$\begin{aligned}&\mathbb{P}(\mu_{\bar{X}} - 0.96\sigma_{\bar{X}} < \bar{X} < \mu_{\bar{X}} + 0.25\sigma_{\bar{X}}) \\ &\quad = \mathbb{P}(-0.96 < Z < 0.25) = \Phi(0.25) - \Phi(-0.96) \approx \boxed{0.4302}.\end{aligned}$$

□

Exercise 8.18. For random samples of size 49 from a large or infinite population, the standard deviation of the sampling distribution of the mean is 3. How large must the sample size be to reduce this standard deviation to 1.5?

Solution. The standard error is $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. With the same population variance,

$$\frac{1.5}{3} = \sqrt{\frac{49}{n}} \iff n = 49\left(\frac{3}{1.5}\right)^2 = \boxed{196}.$$

Thus quadrupling the sample size halves the standard error. □

Exercise 8.19. An automobile battery has mean lifetime 1500 days and standard deviation 250 days. How does the variance of the sample mean change when the sample size is

(a) increased from 49 to 225?

(b) decreased from 625 to 36?

Solution. Independence of sample observations gives $\mathrm{Var}(\bar{X}) = \sigma^2/n = 62500/n$ (days^2).

(a) $\mathrm{Var}(\bar{X}_{49}) = \dfrac{62500}{49} \approx 1275.5102$ and $\mathrm{Var}(\bar{X}_{225}) = \dfrac{62500}{225} \approx \boxed{277.7778}$. The variance decreases by the factor $49/225$.

(b) $\mathrm{Var}(\bar{X}_{625}) = \dfrac{62500}{625} = 100$ and $\mathrm{Var}(\bar{X}_{36}) = \dfrac{62500}{36} \approx \boxed{1736.1111}$. The variance increases by the factor $625/36$.

□

Exercise 8.21. A soft-drink machine dispenses an average of 240 mL per drink, with standard deviation 15 mL. From time to time, a random sample of 40 drinks is taken. The machine is considered satisfactory if the sample mean lies within $\mu_{\bar{X}} \pm 2\sigma_{\bar{X}}$; otherwise it is adjusted. An observed sample mean is 236 mL. Was the decision not to adjust the machine reasonable?

Solution. For a random sample of size 40,

$$\mu_{\bar{X}} = 240, \qquad \sigma_{\bar{X}} = \frac{15}{\sqrt{40}} \approx 2.3717 \text{ mL}.$$

The acceptance interval is

$$\left[240 - \frac{30}{\sqrt{40}},\ 240 + \frac{30}{\sqrt{40}}\right] \approx [235.257, 244.743] \text{ mL}.$$

The observed mean 236 mL lies within the acceptance interval. Therefore no adjustment is required under the specified rule; the sample mean need not equal 240 exactly. □

Exercise 8.25. The time taken to complete a project at a real estate company has mean 18 months and standard deviation 3 months; assume project completion times are approximately normal. Find

(a) the probability that the mean completion time for four independent projects lies between 16 and 19 months.

(b) the value of $\bar{x}$ above which 20% of sample means of size five would fall.

Solution. (a) For $n = 4$, $\bar{X}$ is approximately $\mathcal{N}(18, 9/4)$, with standard error $3/\sqrt{4} = 1.5$. Thus

$$\begin{aligned}\mathbb{P}(16 < \bar{X} < 19) &\approx \Phi\left(\frac{19-18}{1.5}\right) - \Phi\left(\frac{16-18}{1.5}\right) \\ &= \Phi\left(\frac{2}{3}\right) - \Phi\left(-\frac{4}{3}\right) \approx \boxed{0.6563}.\end{aligned}$$

(b) For $n = 5$, the upper-tail probability 0.20 corresponds to the 80th percentile, $z_{0.80} = \Phi^{-1}(0.80) \approx 0.8416$. Therefore

$$\boxed{\bar{x} = 18 + \frac{3}{\sqrt{5}} z_{0.80} \approx 19.13 \text{ months}}.$$

□

Exercise 8.35. An electrical firm makes light bulbs whose lifetimes are approximately normal with mean 800 hours and standard deviation 40 hours. A random sample of 16 bulbs has mean lifetime $\bar{x} = 775$ hours. Does this observation cast doubt on the premise $\mu = 800$? Calculate $\mathbb{P}(\bar{X} \le 775)$ first when $\mu = 800$, and then when $\mu = 760$, keeping the standard deviation 40 hours.

Solution. With $n = 16$, the sampling standard deviation is $40/\sqrt{16} = 10$ hours. Under the assumption $\mu = 800$,

$$\mathbb{P}(\bar{X} \le 775) \approx \Phi\left(\frac{775 - 800}{10}\right) = \Phi(-2.5) \approx \boxed{0.0062}.$$

Under the alternative assumption $\mu = 760$, with the same standard deviation,

$$\mathbb{P}(\bar{X} \le 775) \approx \Phi\left(\frac{775 - 760}{10}\right) = \Phi(1.5) \approx \boxed{0.9332}.$$

A sample mean this low is quite unusual when $\mu = 800$, but is not unusual when $\mu = 760$. The observation therefore casts doubt on the original mean of 800 hours; it does not by itself determine the true population mean. □

Exercise 8.60. Five retail shops are selected at random from a city, and their quoted prices for a home appliance are \$305, \$312, \$296, \$304, and \$307. Find the sample variance of these quotations.

Solution. The sample size is 5, and

$$\sum_{i=1}^{5} x_i = 1524, \qquad \sum_{i=1}^{5} x_i^2 = 464650, \qquad \bar{x} = \frac{1524}{5} = 304.8.$$

Therefore

$$s^2 = \frac{464650 - 1524^2/5}{5 - 1} = \boxed{33.7 \text{ dollars}^2}.$$

The denominator is $n - 1 = 4$, since the quotations constitute a sample. □

Exercise 8.72. A random variable X is normally distributed with mean 20 and variance 9. Determine the smallest positive integer sample size n for which the mean $\bar{X}$ of an independent random sample satisfies

$$\mathbb{P}(19.9 \le \bar{X} \le 20.1) \ge 0.95.$$

Solution. Because X is normal, $\bar{X} \sim \mathcal{N}(20, 9/n)$ exactly. By symmetry and standardization,

$$\mathbb{P}(19.9 \le \bar{X} \le 20.1) = \mathbb{P}\left(-\frac{0.1\sqrt{n}}{3} \le Z \le \frac{0.1\sqrt{n}}{3}\right) = 2\Phi\left(\frac{0.1\sqrt{n}}{3}\right) - 1.$$

The required probability is at least 0.95 precisely when

$$\frac{0.1\sqrt{n}}{3} \ge \Phi^{-1}(0.975) \approx 1.959964, \quad \text{that is,} \quad n \ge \left(\frac{3\Phi^{-1}(0.975)}{0.1}\right)^2 \approx 3457.313.$$

Thus the minimum integer sample size is $\boxed{n = 3458}$. □